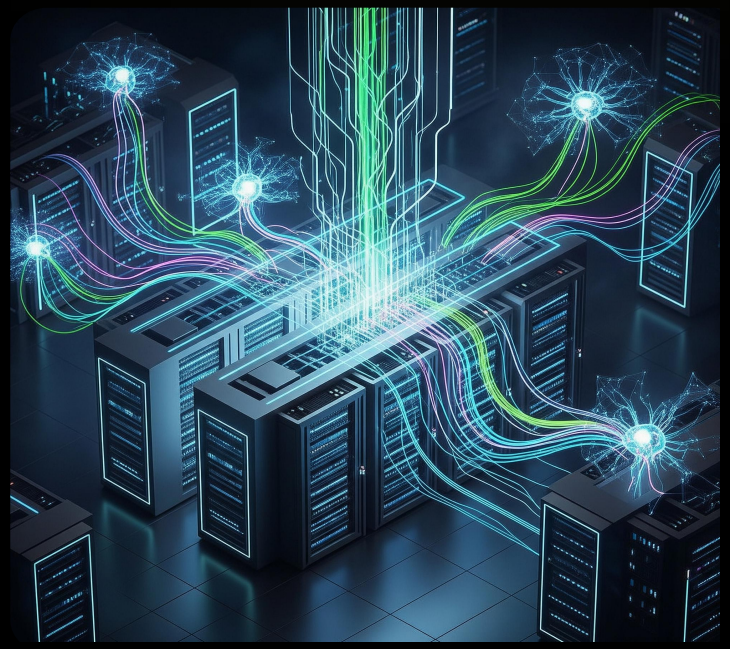
GUEST LECTURE SERIES

The Role of Mass Storage in Storing Training Data for AI Models



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Overview of AI and Training Data

The Data Engine

- AI relies on large datasets for learning
- Training data is essential for model accuracy
- Data quality directly impacts AI performance
- Diverse data types (images, text, video)
- Continuous data collection for relevance

Mass Storage Role

- Handles exabytes of unstructured data
- Ensures 24/7 data accessibility
- Supports high-speed data processing
- Provides necessary scalable solutions
- Enhances data security and integrity

Defining Mass Storage Ecosystems

HDDs & SSDs

Traditional high-capacity mechanical drives vs fast, reliable Solid State Drives.

Examples: Samsung, Seagate, WD

Cloud Storage

Highly scalable, accessible platforms for distributed AI workloads.

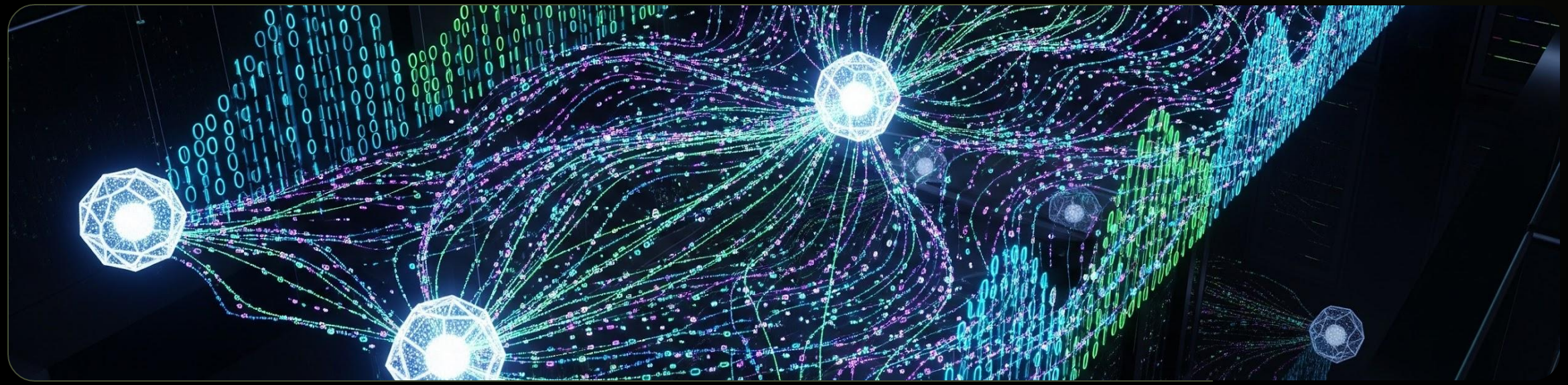
AWS, Google Cloud, Azure

NAS & Tape

Network Attached Storage for teams and Tape for long-term cold archiving.

Synology, QNAP, IBM

The Strategic Value of Data Assets



Role in Model Training

Data enables model learning by providing the patterns required for inference. Without quality, diverse datasets, AI accuracy plateaus.

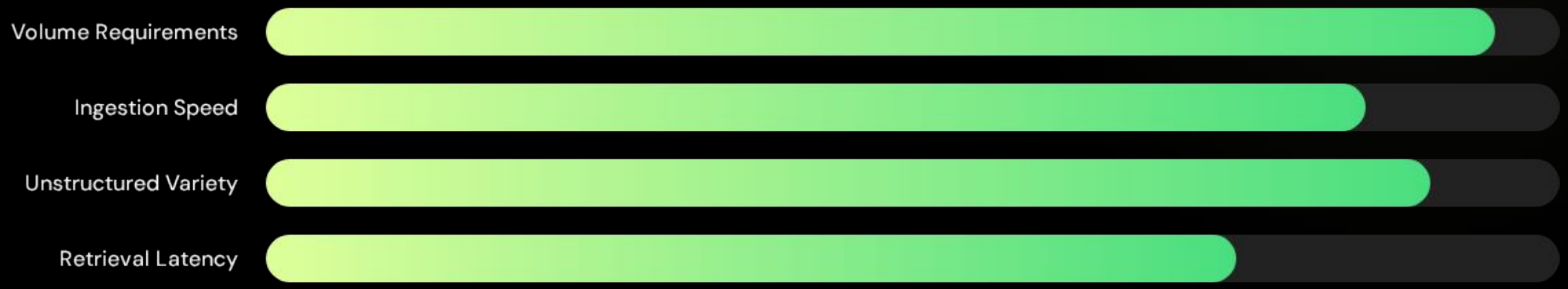
Images
Computer Vision

Text
NLP Models

Audio
Speech Tech

Sensor
IoT Apps

Critical Storage Challenges



Volume & Variety: Managing massive growth of unstructured data requires flexible indexing.

Speed & Accessibility: Low latency is vital for real-time processing and model iteration.

Traditional vs Modern Architecture

Legacy Infrastructure

Focus on capacity over performance.

- → HDDs: High capacity, low cost
- → Tape: Offline durability
- → Manual data management
- → Fixed configurations

AI-Ready Infrastructure

Focus on scalability and efficiency.

- → SSDs/NVMe: Ultra-fast I/O
- → Cloud: Global accessibility
- → Automated data tiering
- → Elastic scalability

File-Based Storage Systems

Hierarchical Structure
Organizes data into traditional files and folders. Highly suitable for structured datasets and legacy integration.

- Easy to use and manage
- Common in traditional IT
- Widely supported by apps
- Ideal for smaller datasets

Technical Specs

Examples: NTFS, ext4, ZFS

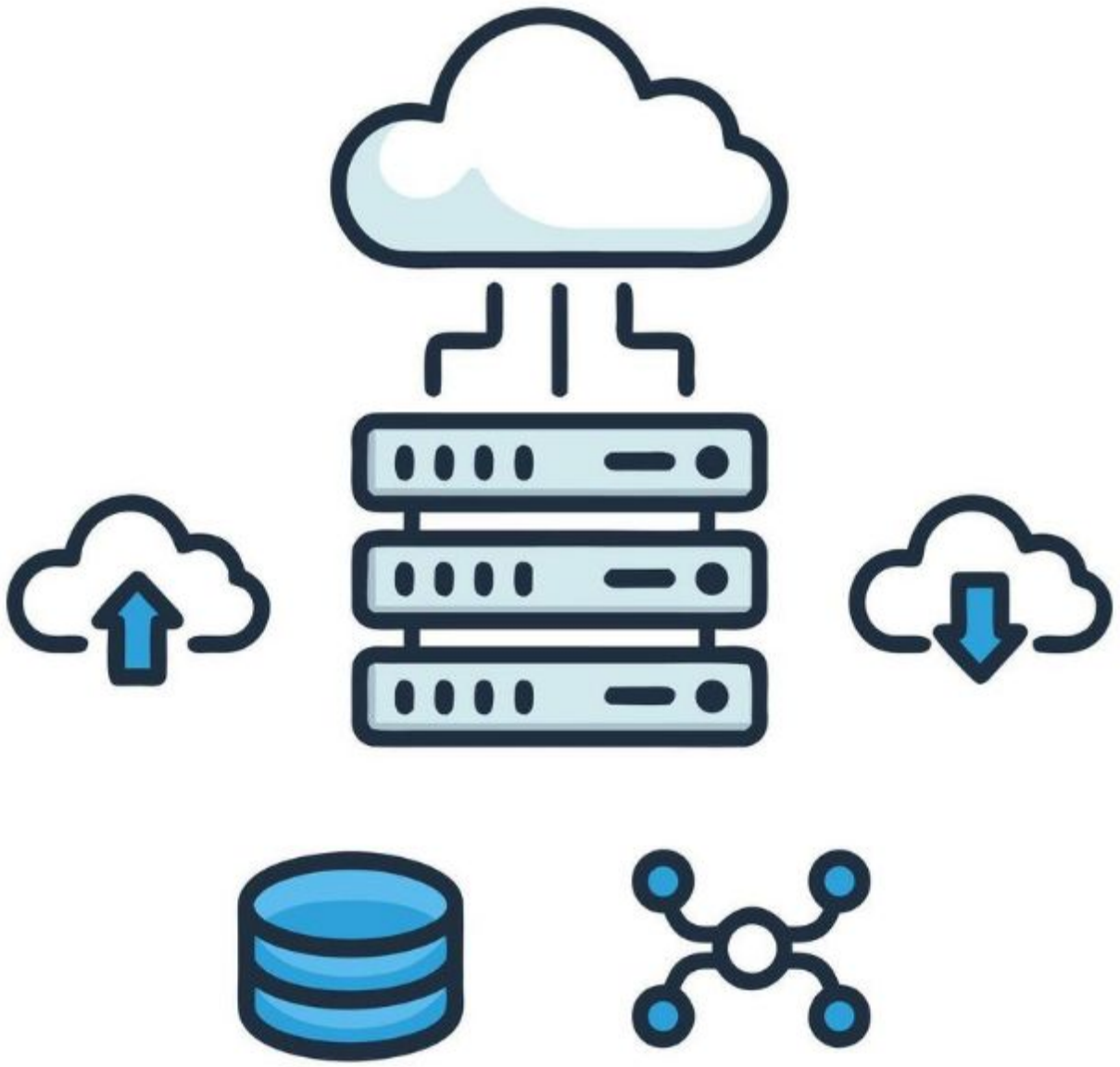
Good for document storage and collaborative environments where file sharing is the primary use case.

Object Storage Architectures

Scale Without Limits
Manages data as discrete objects with unique identifiers and extensive metadata.

- → Exascale scalability for AI
- → Ideal for unstructured data
- → Cost-effective big data storage
- → Seamless cloud integration

Leaders: Amazon S3, Azure Blob, Google Cloud Storage



Strategic Cloud Benefits

AWS

Elastic capacity with S3 and FSx for Lustre integration.

Google Cloud

Highly optimized for TensorFlow and TPU workloads.

Azure

Enterprise-grade security and integration with ONNX.

Pay-as-you-go	**Global Access**	**99.99% Uptime**	**Compliance**
Cost efficiency	Remote teams	Reliability	Security

High-Performance Computing (HPC)

100GB/s

Typical Ingestion Throughput

Modern AI workloads require parallel file systems that stripe data across thousands of drives simultaneously.

Optimized Parallelism

Lustre and GPFS (IBM Spectrum Scale) provide the massive concurrency needed for GPU clusters.

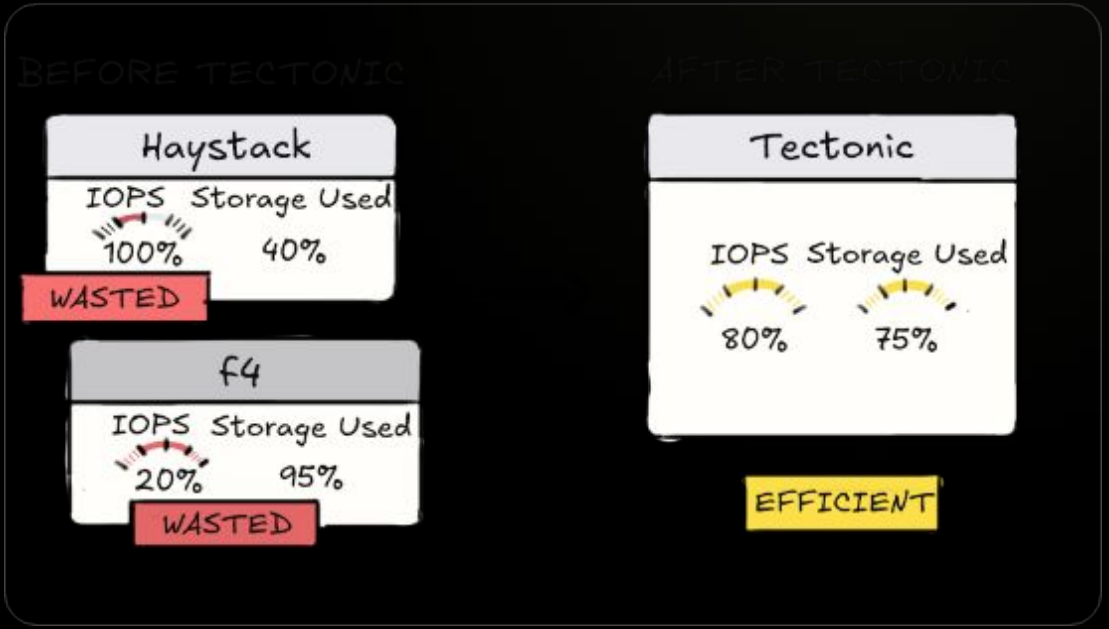
- → Ultra-high I/O throughput
- → Sub-millisecond latency
- → Massive scale-out capacity
- → Direct RDMA support

Case Study: Meta's AI Training

The Challenge

Meta faced exabyte-scale growth and required high-speed access for training Llama models.

- → Exabyte distributed file system (Tectonic)
- → Optimized data pipelines to eliminate stalls
- → Integrated flash tiers for hot data



Key Lesson: Decoupling storage from compute is essential for independent scaling.

Decentralized Cloud Storage

35%

Average Cost Reduction

Decentralized cloud object storage leverages distributed nodes to improve security and lower egress fees.

Implementation Benefits

- Improved data accessibility
- Enhanced security via sharding
- Scalable growth on-demand
- Reduced storage overhead

This model proves that decentralized infrastructure is now viable for production AI.

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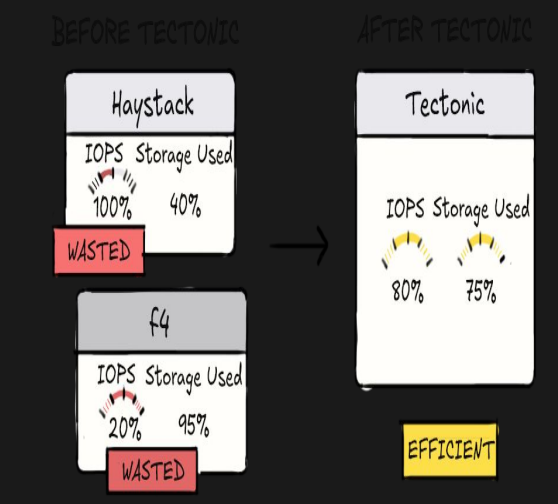

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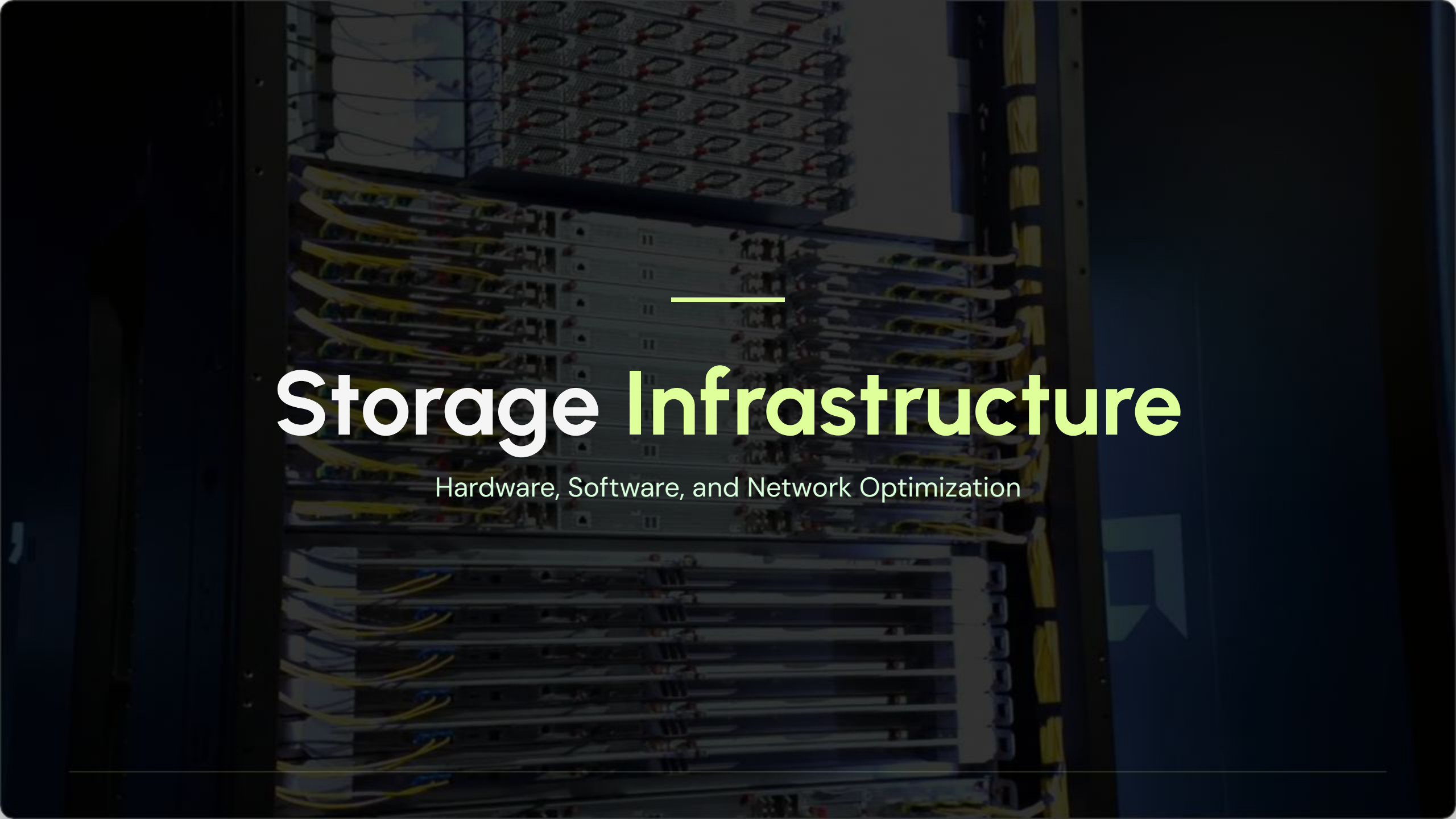
AI-Optimized Features

Feature	Standard Storage	AI-Optimized Storage
I/O Performance	Moderate	Ultra-High / NVMe-based
Latency	10ms+	Sub-millisecond
Data Variety	Primarily Structured	Unstructured & Multimedia
Integration	OS-level	Native AI Framework Support
Scalability	Vertical	Horizontal / Disaggregated



Storage Infrastructure

Hardware, Software, and Network Optimization

Future Roadmap & Trends

NVMe-oF

Fabric-based storage reaching memory-like speeds.

QLC SSDs

High-density flash replacing mechanical HDDs.

AI-Ops

Self-optimizing storage using predictive AI.

CXL 3.0

Compute Express Link for shared memory pools.

"The impact of AI on storage is transforming modules from passive capacity to intelligent elements."

Resilience: Backup & Recovery

Cloud Replication

60%

Primary automated data redundancy across distributed cloud regions.

Offsite Snapshots

25%

Point-in-time recovery copies stored in geographically isolated facilities.

Immutable Archival

15%

WORM storage for long-term compliance and ransomware protection.

Protecting exabytes of data requires automated backup systems and robust disaster recovery plans.

Cybersecurity & Adversarial Defense

Infrastructure Defense

→End-to-end data encryption

→Multi-factor authentication

→Immutable storage snapshots

→Regular vulnerability testing

Adversarial Defense

Protecting models from poisonous data injection and malicious extraction.

Using secure data sources is a primary requirement for AI integrity.

"Data is the fuel of AI, but mass storage is the engine that keeps it moving. In the age of exascale models, infrastructure is the ultimate competitive advantage."

— Final Takeaway, Azizul Azhar Ramli

SCALABILITY • SPEED • SECURITY

Q&A

Thank you for your attention!

Open Floor for Questions and Discussion

ありがとうございます

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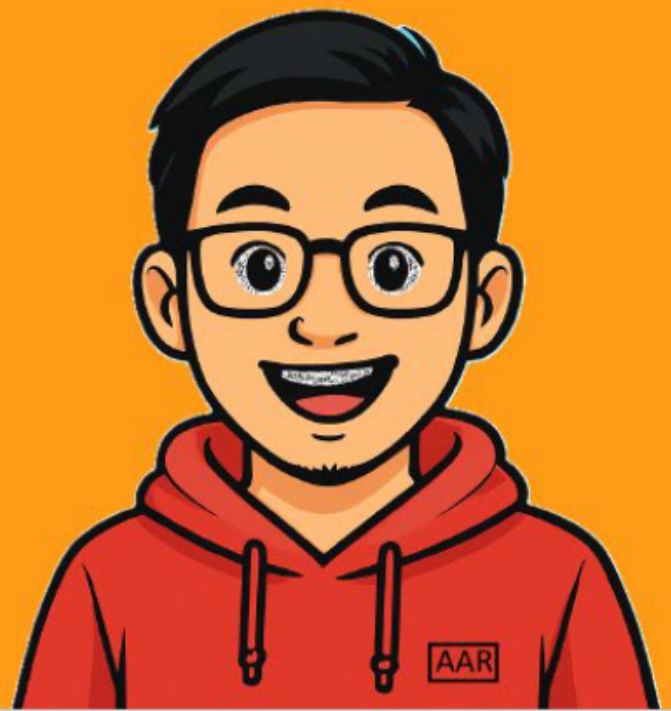
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